

Name: Dallas, Eliana, and Michael

Date: February 5th

Class: P1

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

Explain how this design requirement directly ties to your prior research.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #1 (HIGH Priority): Neutral Buoyancy

- Sits just below the water (1 to 15 m)

Design Requirement #2 (HIGH Priority): Collects microplastics

- This is the whole purpose of the mission, so it is extremely important
- No water collected!!

Design Requirement #3 (HIGH Priority): Measure how fast water is moving in certain areas

- Can observe how microplastics circulate the area
- More data is always good

Design Requirement #4 (HIGH Priority): Multiple ways of collecting microplastics (sediment and water)

- Gives more of an accurate representation of the microplastics in a certain area
- More data is always good

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Design Requirement #5 (HIGH Priority): Moves efficiently

- Easily controlled
- No issues getting it in and out of the water
- This will make conducting the experiment faster and easier

Design Requirement #6 (LOW Priority): Use as little plastic as possible

- Reduce harmful materials
- Makes data more accurate

Design Requirement #7 (LOW Priority): Low cost

- Easily accessible for people to buy

Design Requirement #8 (LOW Priority): Palatable size

- Can be carried and transported by one person
- Not too heavy or too big